

UNIT 12

ACOUSTICS



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SOUND WAVES:

"Sound is a mechanical wave (Longitudinal) that is created by vibrating objects."

Frequencies of sound waves:**1. Audible sound:**

"Sound that could be heard is called audible sound."

Frequency range:

Normal human ear can detect longitudinal waves having frequency between 20 Hz and 20,000 Hz.

Examples:

- Sound of feathers of birds.
- Sound of insects.
- Sound of human vocal cords.
- Sound of musical instruments.

2. Ultrasonic sound:

"Sound whose frequency is more than the maximum range of audible frequency is known as ultrasonic sound."

Examples:

- Bats and whales use echo location that can reach frequency in excess of 100 KHz.

3. Infrasonic sound:

"Sound whose frequency is less than the minimum range of audible frequency is known as infrasonic sound."

Examples:

- Earthquake waves.
- A rhinoceros uses frequencies down to 10 Hz.

SPEED OF SOUND WAVES:**NEWTON'S FORMULA FOR SPEED OF SOUND:**

According to Newton's formula, sound is transmitted through a medium in the form of longitudinal waves. Thus its velocity or speed depends on:

- The elastic modulus 'E' of the medium.
- Density 'ρ' of the medium.

Newton formulated an equation of the velocity of sound, which is known as Newton's formula and it is written as:

$$v = \sqrt{\frac{E}{\rho}} \quad (1)$$

Newton assumed that during the propagation of sound waves, temperature remains constant, i.e., isothermal process is obeyed.

Consider sound waves propagating in air, and due to compression, the pressure 'P' increases to 'P + ΔP' due to which volume 'V' decreases and becomes 'V - ΔV'.

According to Boyle's law for isothermal process:

$$\begin{aligned} PV &= (P + \Delta P)(V - \Delta V) \\ PV &= PV - P\Delta V + \Delta PV - \Delta P\Delta V \\ P\Delta V &= \Delta PV - \Delta P\Delta V \end{aligned}$$

If the change in pressure is small, then the corresponding change in volume is also negligible, hence neglecting $\Delta P \Delta V$,

$$P \Delta V = V \Delta P$$

Or we can write the above equation as:

$$P \frac{\Delta P}{\Delta V} = (\alpha)$$

Where ΔP = Volumetric stress and $\frac{\Delta P}{V}$ = Volumetric strain, and their ratio is equal to E = Bulk modulus.

By substituting equation (a) in Equation (1), it becomes:

$$V = \sqrt{\frac{P}{\rho}}$$

Above equation is called Newton's formula for speed of sound in air. For air [$P = P_{\text{Mercury}} h$], so Newton's formula can be written as:

$$V = \sqrt{\frac{P_{\text{Mercury}} h}{\rho}}$$

At STP = 13.6 g/cm^3 , $g = 981 \text{ cm/s}^2$, and $h = 76 \text{ cm}$, density of air $\rho = 1.293 \times 10^{-3} \text{ g/cm}^3$. By substituting these quantities in Newton's formula for speed of sound in air, it is found that:

$$V = \sqrt{\frac{13.6 \times 981 \times 76}{1.293 \times 10^{-3}}} = 28003.447 \text{ cm/s}$$

Error in Newton's formula:

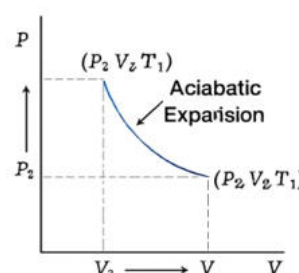
- The experimental value of sound waves in air at STP is found to be 33200 cm/s .
- Therefore, the theoretical value is 15.6% less than the experimental value of speed of sound waves.
- The large difference in theoretical and experimental value cannot be attributed to the experimental errors.
- Newton was unable to explain the error in his formula and the correction was explained by French scientist Laplace.

LAPLACE CORRECTION:

According to Laplace, when sound waves travel through the air, there is compression which rises temperature. Similarly, during rarefactions, temperature falls. The change in temperature, volume, and air pressure is so rapid, and temperature doesn't remain constant. The process is not isothermal but adiabatic.

For an adiabatic process, Boyle's law is given by:

$$PV^\gamma = \text{Constant}$$



If the pressure of the gas changes from P to $P + \Delta P$, resulting in a change of volume from V to $V - \Delta V$, then:

$$PV^\gamma = (P + \Delta P) (V - \Delta V)^\gamma$$

By dividing and multiplying V on the right side:

$$PV^\gamma = (P + \Delta P) \left(V - \frac{\Delta V}{V} \times V \right)^\gamma$$

$$PV^\gamma = (P + \Delta P) \left\{ v \left(1 - \frac{\Delta V}{V} \right) \right\}^\gamma \left(1 - \frac{\Delta V}{V} \right)^\gamma$$

$$PV^\gamma = (P + \Delta P) v^\gamma \left(1 - \frac{\Delta V}{V} \right)^\gamma$$

$$PV^\gamma = (P + \Delta P) v^\gamma \left(1 - \frac{\Delta V}{V} \right)^\gamma$$

By using binomial expansion:

$$P = (P + \Delta P) \left(1 - \gamma \frac{\Delta V}{V} \right)$$

$$P = P - P \gamma \frac{\Delta V}{V} + \Delta P - \Delta P \gamma \frac{\Delta V}{V}$$

Neglecting the term $\Delta P \gamma \frac{\Delta V}{V}$,

$$P = P - P \gamma \frac{\Delta V}{V} + \Delta P$$

$$\Delta P = P \gamma \frac{\Delta V}{V}$$

$$\gamma P = \frac{\Delta P}{\frac{\Delta V}{V}}$$

Thus, we can write the above equation as:

$$\gamma P = \frac{\Delta P}{\frac{\Delta V}{V}}$$

Where ΔP is the volumetric stress and $\frac{\Delta V}{V}$ is the volumetric strain, and their ratio is equal to E = Bulk Modulus.

By substituting this value in Newton's formula for speed of sound,

$$V = \sqrt{\frac{\gamma P}{\rho}}$$

Here γ is the ratio between molar specific heat of gas at constant pressure and volume, and for air, $\gamma = 1.42$.

Hence, according to Laplace correction, the speed of sound in air is given by:

$$V = \sqrt{\frac{1.42 \times 13.6 \times 981 \times 76}{1.293 \times 10 - 3}}$$

$$V = 33369.9 \text{ cm/s} \approx 333 \text{ m/s}$$

The value of speed of sound is in good agreement with the experimental value 332m/s at 0°C.

FACTOR AFFECTING THE SPEED OF SOUND IN AIR:

1. Density of air:

Speed of sound in air is inversely proportional to the square root of the density of air.

2. Moisture (Humidity):

The presence of moisture in air decreases the resultant density of air, which increases the speed of sound in humidity. Hence, the speed of sound in damp (Wet) air is greater than in dry air.

3. Pressure:

According to the general gas equation:

$$PV = nRT \Rightarrow P = \frac{nRT}{v} \text{ and } \rho = \frac{m}{v}$$

Hence, the formula for the speed of sound can be written as:

$$V = \sqrt{\frac{\gamma P}{\rho}} \Rightarrow \sqrt{\frac{\gamma \frac{nRT}{v}}{\frac{m}{v}}} \Rightarrow \sqrt{\frac{\gamma n R T}{m}}$$

Where $\frac{m}{n}$ is the molar specific heat.

Above equation shows that the speed of sound is independent of the pressure of gas (Air).

4. Temperature:

Temperature changes do not affect the speed of sound in solids and liquids significantly.

For gases, the rise and fall of temperature speed of sound changes significantly.

There is an increase of 0.6 m/s in the speed of sound in air for each rise of 1°C temperature in air. Thus, the equation of sound in air is:

$$V_T = 332 + 0.6 \times \Delta T$$

This equation is valid only when we consider the initial value at 0°C.

5. Wind Speed and Direction:

If the speed of wind is V_w , then the speed of sound along the direction of the wind relative to the ground is $(V + V_w)$, and against the direction of the wind, it is $(V - V_w)$.

PRINCIPLE OF SUPERPOSITION:

"When two or more waves overlap, the net disturbance at any point is the sum of the individual disturbances due to each wave."

$$Y = Y_1 + Y_2$$

INTERFERENCE OF SOUND WAVES:

"The principle of superposition leads to extraordinary effects when applied to coherent sound waves, which is called interference of sound waves."

TYPES OF INTERFERENCE OF SOUND WAVES:

1. Constructive interference of sound waves:

"The superposition of the waves that are in phase that is, the crest of one is fall on the crest of the other is called constructive interference."

2. Destructive interference of sound waves:

"The superposition of the waves that are 180° out of phase is called destructive interference."

BEATS:

"The periodic variation in intensity at a given point due to the superposition of two sound waves having nearly the same frequencies, same wave lengths, and same amplitudes and slightly different phases traveling in the same direction are called beats."

Expression for Beat Frequency:

Consider two waves of equal amplitude A_0 traveling through a medium in the same direction having slightly different frequencies. If f_2 be the frequency of one wave, then its

instantaneous displacement Y_1 at the instant t is given by:

$$Y_1 = Y_0 \sin (Kx - \omega t)$$

OR

$$Y_1 = A_0 \cos 2\pi f_1 t$$

Similarly, if f_2 be the frequency of the other wave, then its displacement Y_2 at the instant t is given by:

$$Y_2 = y_0 \sin (kx - \omega t - \phi)$$

OR

$$Y_2 = A_0 \cos 2\pi f_2 t$$

If 'Y' be the resultant displacement caused by the two waves, then:

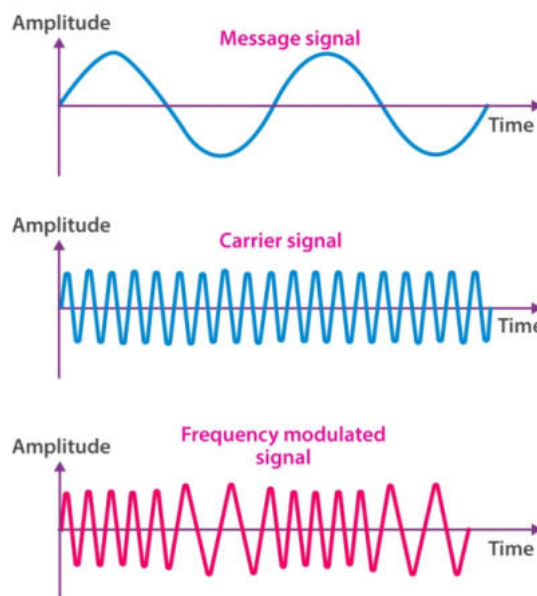
$$Y = Y_1 + Y_2$$

$$Y = (A_0 \cos 2\pi f_1 t) + (A_0 \cos 2\pi f_2 t)$$

$$\therefore \left[\cos A + \cos B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2} \right]$$

$$Y = 2A_0 \cos \frac{2\pi f_1 t - 2\pi f_2 t}{2} \cos \frac{2\pi f_1 t + 2\pi f_2 t}{2}$$

$$Y = 2A_0 \cos 2\pi \frac{(f_1 - f_2)}{2} t \cos 2\pi \frac{(f_1 + f_2)}{2} t$$



In the above equation, $\frac{(f_1 - f_2)}{2}$ is the effective frequency and $2\pi \frac{(f_1 + f_2)}{2}$ is the amplitude A .

$$A = 2A_0 \cos 2\pi \frac{(f_1 - f_2)}{2} t$$

The amplitude will be maximum when:

$$2A_0 \cos 2\pi \frac{(f_1 - f_2)}{2} t \pm 1$$

This means that the variation of frequency $\frac{(f_1 - f_2)}{2}$ occurs twice in one second, so the beat frequency is given by:

$$F_b = 2 \frac{(f_1 - f_2)}{2}$$

$$F_b = (f_1 - f_2)$$

This shows that the number of beats per second will be equal to the difference of frequencies of two waves.

STATIONARY (STANDING) WAVES IN STRING:

"If tightly stretched string is plucked from half of its string, then the disturbance form in the string is called a stationary wave."

Nodes:

"The points where amplitude is zero are called nodes."

Antinodes:

"The points where the amplitude is maximum are called antinodes."

MODES OF VIBRATION IN STRETCHED STRING:

Let us consider a string of length L stretched between two fixed points. When plucked

from the midpoint and then released, a standing wave is formed as shown in the figure.

First Harmonic or Fundamental Frequency:

If the frequency and wavelength of the wave are f_1 and λ_1 , respectively, then it is clear from the figure that:

$$L = \frac{\lambda_1}{2}$$

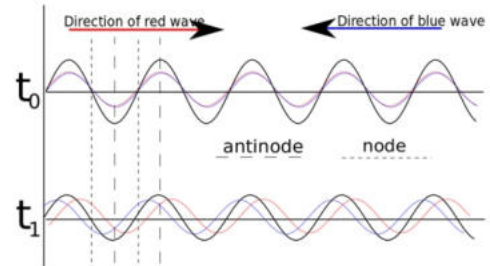
$$\lambda_1 = 2L$$

Since,

$$V = f_1 \lambda_1$$

$$f_1 = \frac{V}{\lambda_1}$$

$$f_1 = \frac{V}{2L}$$



The frequency f_1 so obtained is called the fundamental frequency or first harmonic of the standing waves.

If the tension in the string is T and the linear density of the string is μ , then:

$$V = \sqrt{\frac{T}{\mu}}$$

And

$$f_1 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Second Harmonic (First Overtone):

When the loop is touched by a light feather at its midpoint, then the string starts vibrating in two loops. In this case, the wavelength and frequency are $f_2 \lambda_2$, respectively.

According to the condition:

$$\frac{\lambda_2}{2} + \frac{\lambda_2}{2} = L$$

$$\frac{\lambda_2}{2} = L$$

$$\frac{2L}{2} = \lambda_2$$

Since,

$$V = f_2 \lambda_2$$

$$f_2 = \frac{V}{\lambda_2}$$

Therefore,

$$f_2 = \frac{V}{\frac{2L}{2}}$$

$$f_2 = \frac{2V}{2L}$$

$$f_2 = 2 \frac{V}{2L}$$

Where, $\frac{v}{2L} f_1$

$$f_2 = 2f_1$$

Third harmonic or second overtone:

Similarly, if the string vibrates in three loops then,

$$\frac{\lambda_3}{2} + \frac{\lambda_3}{2} + \frac{\lambda_3}{2} = L$$

$$\frac{3\lambda_3}{2} = L$$

$$\frac{2L}{3} = \lambda_3$$

Since,

$$v = f_3 \lambda_3$$

$$f_3 = \frac{v}{\lambda_3}$$

Therefore,

$$f_3 = \frac{v}{\frac{2L}{3}}$$

$$f_3 = \frac{3v}{2L}$$

$$f_3 = 3 \frac{v}{2L}$$

Where, $\frac{v}{2L} = f_1$

$$f_3 = 3f_1$$

For 'n' loops:

By generalizing the results of f_1 , f_2 and f_3 , if string vibrates in 'n' loops then,

$$f_n = nf_1$$

Where $n = 1, 2, 3, 4, \dots$

ORGAN PIPE:

An organ pipe is a wind instrument. It consists of a long tube in which air is forced from one end and sound is produced by mean of vibrating air column.

Types of organ pipe:

1. Closed pipe (One end of organ pipe is closed).
2. Open pipe (Both end of organ pipe is open).

Open pipe (both end of organ pipe is open):

If the pipe is open at both ends, at each end the column of air inside the pipe communicates with the outside air so the pressure at ends can't deviate much from atmospheric pressure. The open ends are therefore pressure nodes.

There are also displacement anti-nodes, element of air vibrate back and forth with maximum amplitude at the ends.

Since nodes and anti-nodes alternate with equal spacing i.e. $\lambda/4$.

Hence the equation for frequencies is same as stationary waves i.e. $f_n = nf_1$

Closed pipe (One end of organ pipe is open):

If the pipe is closed at one end, the closed end is the pressure anti-node; the air at closed end meets a rigid boundary, so there is no limit that how far the pressure can deviate from atmospheric pressure.

The closed end is also a displacement node since the air near it cannot move beyond that rigid surface.

Since, it is mentioned earlier that the distance between a node and antinode shall always be $\lambda/4$ for fundamental mode, hence:

First harmonic:

$$L = \lambda/4$$

Or

$$\lambda = 4L$$

Which is twice as large as the wavelength ($2L$) of fundamental mode of vibration in a pipe of same length, open at both ends,

$$f_1 = \frac{v}{\lambda_1} \Rightarrow f_1 = \frac{v}{4L}$$

Second harmonic:

If the air column vibrates for second harmonic or first overtone then,

$$L = \frac{\lambda_2}{2} + \frac{\lambda_2}{4} \Rightarrow \frac{3\lambda_2}{4}$$

$$\lambda_2 = \frac{4L}{3}$$

Since,

$$f_2 = \frac{v}{\lambda_2}$$

$$f_2 = \frac{v}{\frac{4L}{3}}$$

$$f_2 = \frac{3v}{4L}$$

$$f_2 = 3\left(\frac{v}{4L}\right)$$

$$f_2 = 3f_1$$

Third harmonic:

If the air column vibrates for third harmonic then,

$$L = \frac{\lambda_3}{2} + \frac{\lambda_3}{2} \Rightarrow \frac{\lambda_3}{2} \Rightarrow \frac{5\lambda_3}{4}$$

$$\lambda_3 = \frac{4L}{5}$$

Since,

$$f_3 = \frac{v}{\lambda_3}$$

$$f_3 = \frac{v}{\frac{4L}{5}}$$

$$f_3 = \frac{5v}{4L}$$

$$f_3 = 5\left(\frac{v}{4L}\right)$$

$$f_3 = 5f_1$$

For 'n' harmonics the standing wave frequencies for closed pipe at one end are odd integral multiples of the fundamental frequencies,

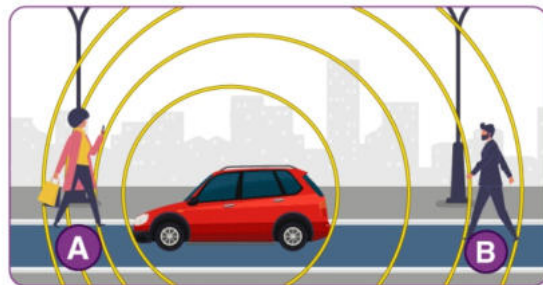
$$f_n = nf_1 \text{ (Where } n = 1, 3, 5, \dots)$$

DOPPLER'S EFFECT:

"The change in the frequency of the sound caused by relative motion of either the source of sound or the listener is called Doppler's effect."

Doppler's effect can be explained by considering three possibilities:

1. When the source is moving.
2. When the listener is moving.
3. When both the source and listeners are moving.



CASE #1 (A): WHEN SOURCE IS MOVING TOWARDS STATIONARY LISTENER:

If the source is moving at velocity v_s towards a stationary listener and sounding a tone of frequency f_s . The wave crest detected by the stationary listener are close together because the source is moving in the direction of outgoing waves resulting in the shortening of wave length.

So the wavelength of sound waves will be decreased by the amount of the distance travelled by source.

$$\lambda' = \lambda - \lambda_s$$

Where,

λ = wavelength of sound waves

λ_s = wavelength/distance travelled by source

But,

$$\lambda = v T_s, \lambda_s = v_s \cdot T_s \text{ and } \lambda' = v T'$$

$$v T' = T_s (v - v_s)$$

$$T' = \frac{T_s (v - v_s)}{v}$$

$$\frac{1}{T'} = \frac{v}{T_s (v - v_s)}$$

$$f_1 = \left(\frac{v}{(v - v_s)} \right) f_s$$

This shows that the observed frequency is higher than the source frequency when the source moves in the same direction as the wave towards the observer.

CASE #1 (B): WHEN SOURCE IS MOVING AWAY FROM THE STATIONARY LISTENER:

If the source is moving at velocity v_s away from the stationary listener and sounding a tone of frequency f_s . The wave crest detected by the stationary listener is wide spread because the source is moving away from the direction of incoming waves resulting in the increase of wavelength. So the wavelength of sound waves will be increased by the amount of the distance travelled by source.

$$\lambda' = \lambda + \lambda_0$$

Where,

λ_s = wavelength of sound waves

λ_0 = wavelength/distance travelled by source

λ' = apparent wavelength

But,

$\lambda = vT_s$ $\lambda_s = v_s T_s$ and $\lambda' = vT'$

$$vT' = vT_s + v_s T_s$$

$$vT' = T_s (v + v_s)$$

$$T' = \frac{T_s (v + v_s)}{v}$$

$$\frac{1}{T'} = \frac{v}{T_s (v + v_s)}$$

$$f' = \frac{v}{(v + v_s)} f_s$$

This shows that the observed frequency is less than the source frequency, when the source moves away from the stationary observer.

CASE #2 (A): WHEN OBSERVER IS MOVING TOWARDS STATIONARY SOURCE:

Suppose the listener is moving towards a stationary source of sound. The speed of the listener is v_0 . Let the source emits sound waves of frequency f_s , speed v and wavelength λ .

We know that,

$$\lambda' = \lambda - \lambda_s$$

Where,

λ_s = wavelength of sound waves

λ_0 = wavelength / distance travelled by observer

λ' = apparent wavelength

But,

$\lambda_0 = v_0 T'$ $\lambda_s = v T_s$ and $\lambda' = v T'$

$$vT' = vT_s - v_0 T$$

$$vT' = v_0 T - vT_s$$

$$T' (v + v_0) = vT_s$$

$$T' = \frac{vT_s}{(v + v_0)}$$

$$\frac{1}{T'} = \frac{v + v_0}{vT_s}$$

$$f' = \frac{v + v_0}{v} f_s$$

This shows that the observed frequency is higher than the source frequency.

CASE #2 (B): WHEN OBSERVER IS MOVING AWAY FROM THE STATIONARY SOURCE:

Suppose the listener is moving away from the stationary source of sound.

The speed of the listener is v_0 .

Let the source emits sound waves of frequency f_s , speed v , and wavelength λ .

We know that,

$$\lambda' = \lambda_s + \lambda_0$$

Where,

λ_s = wavelength of sound waves

λ_0 = wavelength / distance travelled by observer

λ' = apparent wavelength

But,

$$\lambda_0 = v_0 T' \lambda_s = v T_s \text{ and } \lambda' = v T'$$

$$v T' = v T_s - v_0 T$$

$$v T' = v_0 T - v T_s$$

$$T' (v + v_0) = v T_s$$

$$T' = \frac{v T_s}{(v + v_0)}$$

$$\frac{1}{T'} = \frac{v + v_0}{v T_s}$$

$$f' = \frac{v + v_0}{v} f_s$$

This shows that the observed frequency is less than the source frequency.

CASE #03 (A): WHEN SOURCE AND LISTENERS ARE MOVING TOWARDS EACH OTHER:

If the listener is moving with velocity v_0 towards the source of sound, then frequency changes from f to f' and is given by:

$$f' = \left(\frac{v + v_0}{v} \right) f_s \dots \dots \dots (a)$$

If the source of sound is also moving with velocity v_s towards the listener, then frequency further changes from f' to f'' and is given by:

$$f'' = \left(\frac{v}{v - v_s} \right) f' \dots \dots \dots (b)$$

Putting the value of f' in equation (b) from equation (a),

$$f'' = \left(\frac{v}{v - v_s} \right) \left(\frac{v + v_0}{v} \right) f_s$$

$$f'' = \left(\frac{v + v_0}{v - v_s} \right) f_s$$

CASE #03 (B) WHEN SOURCE AND LISTENERS ARE MOVING AWAY FROM EACH OTHER:

If the listener is moving with velocity v_0 away from the source of sound, then frequency changes from f to f' and is given by:

$$f' = \left(\frac{v - v_0}{v} \right) f_s \dots \dots \dots (a)$$

If the source of sound is also moving with velocity v_s away from listener, then frequency further changes from f' to f'' and is given by:

$$f'' = \left(\frac{v}{v + v_s} \right) f' \dots \dots \dots (b)$$

Putting the value of f' in equation (b) from equation (a),

$$f'' = \left(\frac{v}{v + v_s} \right) \left(\frac{v - v_0}{v} \right) f_s$$

$$f'' = \left(\frac{v - v_0}{v + v_s} \right) f_s$$

WHEN THE VELOCITY OF SOURCE IS EQUAL TO THE VELOCITY OF SOUND:

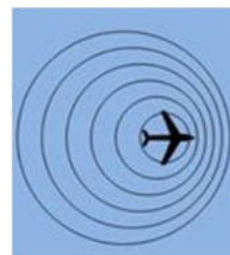
When the velocity of source is equal to velocity of sound i.e. $v_s = v_v$, $v_s = v$ then source and sound waves interact with each other and high pressure is formed. As the process continues the pile region extends further and further from the source in the direction perpendicular to its motion due to which a thick wave front or sheet is formed. This sheet exerts tremendous pressure on the source.

WHEN VELOCITY OF SOURCE IS GREATER THAN THE VELOCITY OF SOUND:

When velocity of source exceeds the wave speed i.e. $v_s > v$ the source runs ahead of outgoing waves and in such a case the wave piles up and produces the shape of a cone with the moving object at its apex.

Along the cone shaped wave front, that is produced by supersonic object (Object whose speed is greater than speed of sound), the air is highly compressed. This moving sheet of high pressure air is called shock waves.

This shock wave will result in loud explosion of sonic boom which can damage buildings if the source is near to it.

Moving Source**APPLICATIONS OF DOPPLER'S EFFECT:****• Echolocation: [Communication]**

Echolocation is a valuable navigation tool for various life forms on our planet. While oil birds and cave swiftlets use audible sound waves for echolocation (detectable by humans), dolphins, whales, and most bats utilize ultrasound (20 Hz to 200 KHz) instead.

Bats and dolphins gain an advantage by sensing the Doppler shift between emitted and reflected waves, allowing them to locate and track fast moving prey effectively.

• Astronomy: [Communication]

The Doppler Effect for light is vital in ASTRONOMY. Analyzing light emitted by elements in distant stars reveals wavelength shifts compared to the same element's light on Earth. These shifts, known as red shifts, indicate motion of the stars.

Redshift observations have provided evidence for expanding universe cosmological theories, suggesting the universe evolved from a great explosion billions of years ago in a relatively small region of space.

• SONAR (Sound Navigation And Ranging): [Communication]

SONAR devices emit ultrasonic pulses to determine distances. The time delay between pulse emission and reflection helps measure seafloor distance.

For low-range submarine detection, the Navy uses low-frequency active (LFA) sonar with audible sound (100 to 500 Hz) instead of ultrasonic.

Air guns are used to study the earth's interior structure and locate oil underground.

• VOR (Very High Frequency Omnidirectional Range): [Communication]

VOR is a guiding system at airports that directs incoming aircraft to the airport's location. Modern airports like Quaid-e-Azam International Airport Karachi and Islamabad Airport use Doppler VOR systems. The electromagnetic signal used operates in the VHF range (30 MHz–300 MHz).

• RADARS (Radio Detection and Ranging): [Communication]

RADAR serve civil and military purposes at airports and air bases, detecting aircraft presence by evaluating range. Doppler radar, based on the Doppler's effect, is used to determine aircraft speed and direction.

It is also crucial in weather forecasting showing storm location and wind velocity.

Radar Speed Traps or Speed (Radar) guns use by traffic police to measure the speed of an automobile or use in Cricket matches to measure the speed of a ball delivered by the bowler. A reflected electromagnetic signal is received from the automobile by the radar gun.

• Cardiovascular Problems: [Medical]

Ultrasound is also used to examine organs such as heart, liver, gallbladder, kidneys, breasts and eyes, and to locate tumors. It can be used to diagnose various heart conditions and to assess damage after a heart attack.

Ultrasound can show movements, so it is used to assess heart valve function and to monitor blood flow in large blood vessels. Since, ultrasound provides real-time images. It is sometimes used to guide procedures such as biopsies, in which a needle is used to take a sample from an organ or tumor for testing.

• Doppler ultrasound: [Medical]

It is a newer technique that is used to examine blood flow. It can help reveal blockage to blood flow, show the formation of plaque in arteries, and provide detail information on the heartbeat of the patient. In ultrasonic imaging, ultrasound waves ($> 20,000$ Hz) rather than sound waves of audible frequencies (20 Hz to 20,000 Hz) are used. Waves with small wavelengths diffract less around the same obstacle than do waves with larger wavelengths. The frequencies used in imaging are typically in the range of 1 to 15 MHz, which means that the wavelengths in human tissue are in the range of 0.1 to 1.5 mm. As a comparison if sound waves at 15 KHz were used, the wavelength inside the body would be 10 cm. Higher frequencies give better resolution but at expense of less penetration; sound waves are absorbed within a less penetration; sound waves are absorbed within a distance of 500λ in tissues.